SEVENTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
Nov2024
SUBJECT: Electronic System Design(ECE-4072)
TIME: 3 HOURS MAX. MARKS: 50

Q1A. Discuss the general criteria and guidelines in product design and give the objectives of Aesthetics and ergonomics while designing a electronic product. Given the rising adoption of electric vehicles (EVs) in India and the push for renewable energy, design a Battery Management System (BMS) with flow chart or block diagram. It should include provisions for remote monitoring through IoT integration and feature a reliable thermal management solution. (4)

Q1B. Design a laboratory product based regulated power supply unit operating on 230V, 50Hz AC signal giving an output voltage of ±12V and output current of 1A. The supply voltage variation can be taken as ±15%. The load regulation should be more than 0.5% and output ripple less than 0.2 %. (3)

Q1C. Design a Fish bone diagram for identifying system failure of any electronic product and do the biostatistical data analysis for reasoning the cause for system failure . (3)

Q2A. Discuss the working of Instrumentation Amplifier with neat circuit Diagram. Derive the equation for its output voltage. (4)

Q2B. Develop a microcomputer based system to sense 4 analogous input signals and produce 4 analogous outputs. The inputs come from the sensors that produce useful signals with a bandwidth of 1KHz which may pickup high frequency noise. The output signals are to drive actuators which may be effected with high frequency signals with the maximum operating band width of 100Hz with signal accuracy of at least 1% . Calculate the sampling rate and the conversion time of the ADC. (3)

Q2C. Discuss PSoC device with an example. (3)

Q3A. Discuss different modes of heat transfer defining newtons law of cooling with its equation. (4)

Q3B. A 15 cm × 20 cm circuit board houses 120 closely spaced 0.12 W logic chips. Calculate the amount of heat dissipated in 10 h and the heat flux on the surface of the circuit board . (3)

Q3C. Discuss the design concept of super capacitors with its applications. (3)

Q4A. Discuss different types of soldering mechanisms. Draw the lead -tin phase diagram indicating the solidification values for perfect soldering. (4)

Q4B. Develop industrial flow chart for manufacturing four layer PCB used in mobile phones. Explain suitable packaging technics used in the semiconductor Industry to increase the reliability of the components and printed circuit board. (3)

Q4C. Discuss differential charge measurement using OPAMP for voltage to current converter circuit. (3)

Q5A. Discuss any one type of input and output device with diagram for any electronic product of your choice and give its features (4)

Q5B. Discuss the effect of noises and electromagnetic interference in a Electronic system. Give suitable methods for its elimination with necessary circuit diagram. (3)

Q5C. Discuss any one type of power amplifier circuits with its application. Compare its performance with respect to the elimination of even harmonics (3)